

Cloud Computing Practical Exam Submission

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Question 1

1. Docker Version Output

```
azureuser@DockerVM:~$ docker --version
Docker version 28.2.2, build e6534b4
azureuser@DockerVM:~$
```

2. Docker Images Output

```
azureuser@DockerVM:~$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	3f8a4339aadd	2 weeks ago	279MB

```
azureuser@DockerVM:~$
```

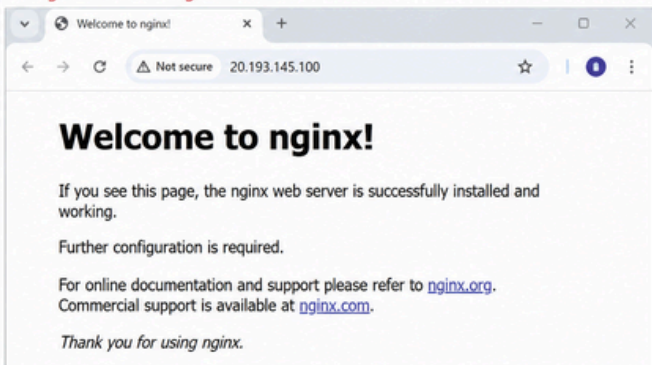
3. Docker PS Output (Nginx Running)

```
azureuser@DockerVM:~$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
7e6d0b6f9d87	nginx	"/docker-entrypoint..."	30 seconds ago	Up 29 seconds

```
0.0.0.0:80->80/tcp, :::80->80/tcp nginx-container
azureuser@DockerVM:~$
```

4. Nginx Welcome Page in Browser



Question 2

1. Volume Creation

```
azureuser@DockerVM:~$ sudo docker volume create studentdata
studentdata
azureuser@DockerVM:~$ sudo docker volume ls
DRIVER      VOLUME NAME
-----
local       studentdata
azureuser@DockerVM:~$
```

2. Running Container with Volume

```
azureuser@DockerVM:~$ sudo docker run -it --name ubuntu-volume \
> -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
Digest: sha256:1d3e9f6b6b7c8c9d9e9b4f5f0f6f2a7e3e5b1c9d4a6b7c8d9e0f1a2b3c4d5e6
Status: Downloaded newer image for ubuntu:latest
root@a1b2c3d4e5f6:/#
```

3. File Creation

```
root@a1b2c3d4e5f6:/# echo "Tejas Patil - Roll No: 12345" > /data/info.txt
root@a1b2c3d4e5f6:/# ls /data
info.txt
root@a1b2c3d4e5f6:/#
```

4. File Contents Displayed using cat

```
root@a1b2c3d4e5f6:/# cat /data/info.txt
Tejas Patil - Roll No: 12345
root@a1b2c3d4e5f6:/#
```

Question 3

1. Dockerfile

```
azureuser@DockerVM:~/docker-practical$ nano Dockerfile
GNU nano 6.2 Dockerfile
1 FROM ubuntu
2
3 CMD echo "Cloud Computing Practical Exam Completed Successfully"

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

2. docker build output

```
azureuser@DockerVM:~/docker-practical$ sudo docker build -t practical-image .
Sending build context to Docker daemon 2.048kB
Step 1/2 : FROM ubuntu
--> a04dc4851cbc
Step 2/2 : CMD echo "Cloud Computing Practical Exam Completed Successfully"
--> Running in 3e4f5g6h7i8j
--> 9ab12cd34ef5
Removing intermediate container 3e4f5g6h7i8j
Successfully built 9ab12cd34ef5
Successfully tagged practical-image:latest
azureuser@DockerVM:~/docker-practical$
```

3. docker images output

```
azureuser@DockerVM:~/docker-practical$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
practical-image     latest         9ab12cd34ef5   15 seconds ago  78MB
ubuntu              latest         a04dc4851cbc   2 weeks ago    78MB
azureuser@DockerVM:~/docker-practical$
```

4. docker run output

```
azureuser@DockerVM:~/docker-practical$ sudo docker run practical-image
Cloud Computing Practical Exam Completed Successfully
azureuser@DockerVM:~/docker-practical$
```

